DIVISOR SUMS AND A COUNTEREXAMPLE TO ERDŐS PROBLEM 859

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Abstract. For a positive integer t, let d_t be the natural density of integers having a set of distinct positive divisors whose sum is t. We prove

$$\liminf_{t\to\infty} \frac{-\log d_t}{\log\log t} = \delta, \qquad (\log t)^{\delta} d_t \longrightarrow 0, \qquad \delta = 1 - \frac{1+\log\log 2}{\log 2}.$$

It follows that d_t is not asymptotic to $c/(\log t)^p$ for any $c, p > 0$, disproving the assertion in Erdős problem 859. The lower bound combines a tilted finite prime model, a proper divisor cube, logarithmic support enlarged by pairs of primes, and control of small Fourier phases. A practical divisor converts bounded-coefficient representations into sums of distinct divisors. Harmonic averaging over targets permits a single change of measure and recovers the critical exponent. An upper bound for divisor intervals gives a vanishing factor at that exponent. A companion single-file Lean proof verifies the negation of the pinned Formal Conjectures statement using only the standard logical axioms.

1. Introduction

For $t \geq 1$, write

$$\mathcal{D}_t = \left\{ n \geq 1 : \ t = \sum_{d\in E} d \text{ for some } E \subseteq \{d : d \mid n\} \right\}.$$

The summands are positive and distinct. Membership in $\mathcal{D}_t$ depends only on the divisibility of n by the integers at most t. Consequently $\mathcal{D}_t$ is periodic, and its natural density d_t exists and is positive. Erdős problem 859 asks whether there are positive constants c and p for which

$$d_t \sim \frac{c}{(\log t)^p} \qquad (t \to \infty). \tag{1.1}$$

We use the formulation recorded in the Formal Conjectures library [3], which refers to Bloom's catalogue [1].

Put

$$a = \log 2, \qquad \lambda_* = a^{-1}, \qquad \delta = 1 - \frac{1+\log a}{a} = 0.0860713320\ldots. \tag{1.2}$$

The same exponent occurs in the distribution of divisors in intervals [2]. Our result shows that the exponent governing the largest values of d_t does not support a pure logarithmic power asymptotic.

Theorem 1.1. *The densities d_t satisfy*

$$\liminf_{t\to\infty} \frac{-\log d_t}{\log\log t} = \delta, \qquad \lim_{t\to\infty} (\log t)^{\delta} d_t = 0. \tag{1.3}$$

Corollary 1.2. *There are no positive constants c, p satisfying* (1.1).

Indeed, the proposed asymptotic would make the first limit in (1.3) equal to p, forcing $p = \delta$. The second assertion would then contradict $(\log t)^{\delta} d_t \to c > 0$. A proof that also handles the precise formal statement appears in Section 5.

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The lower-limit statement is deliberately weaker than an asymptotic formula for d_t. It supplies arbitrarily large targets with the critical logarithmic exponent. It does not assert that all sufficiently large targets have the same exponent, or identify the missing slowly varying factor.

1.1. **The construction.** The lower bound is proved in a finite independent model of the primes dividing a random integer. Under a tilt $\lambda > \lambda_*$, selecting a prime p has probability $\lambda/(p+\lambda-1)$. A block construction produces a core of about k primes with probability

$$\exp\{-k[J(\lambda)+o(1)]\}, \qquad J(\lambda) = \lambda\log 2 - 1 - \log(\lambda\log 2).$$

Its divisors form a proper signed cube at prescribed scales. At the same cost, the selected primes have subset logarithms occupying positive measure in a long interval.

Positive measure alone does not provide usable integer representations. We enlarge the logarithmic support using successive pairs of primes. Choosing one of the two primes implements their logarithmic difference, up to a common translation. After an initial covering stage, a quadratically improving bound on the missing measure supplies a fine mesh. A later prime window is chosen from the previously observed configuration. Its selected primes eliminate all small Fourier phases outside a short arc. Exact conditioning justifies this adaptive choice.

The mesh propagates the phase estimate to actual divisors below a reserve divisor r. Fourier inversion supplies a residue representation with bounded coefficients; the reserve divisor fills an interval of targets. An independent practical divisor makes the summands distinct. The construction represents many targets in harmonic measure. Averaging first, and applying Hölder's inequality only once, preserves the critical exponent under the change back to the original prime law.

The upper bound follows a different route. A finite divisor-support estimate, an ordered-simplex bound, and a decomposition into squarefree and powerful parts give a logarithmic saving for a fixed-ratio divisor interval. A dyadic reduction transfers this saving to d_t.

1.2. **Relation to the formal proof.** The accompanying Lean development proves both estimates in (1.3) and the exact negation of the formalized conjecture. The paper follows the completed finite construction. In particular, its lower bound does not take an infinite random-series absolute-continuity theorem or an asymptotic count of practical numbers as an unproved premise. Related results of Peres, Simon, and Solomyak [5] and Weingartner [6] provide context for the roles of logarithmic support and practical divisors. The formal development instead proves the finite estimates it needs. Its prime-reciprocal estimate adapts an existing Lean proof from the PNT+ project [4].

Section 2 fixes the density and finite probability model. The upper estimate and lower construction follow, with the analytic conclusion in Section 5. Section 6 records the formal target and verification evidence.

All logarithms are natural. An interval's measure is its Lebesgue measure. Constants in $O(\cdot)$ may depend on fixed parameters explicitly held fixed before the limit. A quantity described as having zero exponential cost has logarithm $o(k)$ on the specified sequence of construction parameters.

2. Divisor-sum densities

All logarithms are natural. For an integer $n \geq 1$, write

$$\mathcal{D}(n) = \{d \geq 1 : d \mid n\}, \qquad \Sigma(n) = \left\{\sum_{d\in S} d : S \subseteq \mathcal{D}(n)\right\}.$$

The set S is a set, so a divisor can occur at most once. We include the empty sum. For an integer $t \geq 0$, put

$$\mathcal{D}_t = \{n \geq 1 : t \in \Sigma(n)\}.$$

We write $d(t) = d_t$. Our density convention is

$$d(t) = \lim_{x\to\infty} \frac{1}{x}\#\{n \leq x : n \in \mathcal{D}_t\},$$

when this limit exists. The formalization counts natural numbers below a natural cutoff, including zero. Changing the initial point or one endpoint changes the counting function by at most a constant and hence does not change this density.

Proposition 2.1 (Periodicity and positivity)**.** *For each integer $t \geq 0$, membership in $\mathcal{D}_t$ is periodic on the positive integers with period $t!$. In particular, $d(t)$ exists, is rational, and satisfies*

$$d(t) = \frac{1}{t!}\#\{1 \leq a \leq t! : a \in \mathcal{D}_t\}, \qquad \frac{1}{t!} \leq d(t) \leq 1. \tag{2.1}$$

For $t = 0$ one has $d(0) = 1$.

Proof. The case $t = 0$ follows from the empty sum, so suppose that $t \geq 1$. If $t = \sum_{d\in S} d$ with $S \subseteq \mathcal{D}(n)$, then every $d \in S$ satisfies $1 \leq d \leq t$. Consequently $d \mid t!$. If $n, m \geq 1$ and $n \equiv m \pmod{t!}$, divisibility of n by each such d is equivalent to divisibility of m by d. The same set S therefore represents t using divisors of m. Reversing the roles of n and m proves the periodicity assertion.

Let A be the number of successful residue classes in one period of length $t!$. Splitting a counting interval into full periods and a remainder gives

$$\#\{n \leq x : n \in \mathcal{D}_t\} = \frac{A}{t!}x + O(t!).$$

This proves the density formula. The residue class of $t!$ is successful: $t \mid t!$, so the one-element set $\{t\}$ represents t. Thus $A \geq 1$, giving the positive lower bound. The upper bound is immediate from $A \leq t!$. □

The proof identifies a canonical density without appealing to any unproved assertion in the statement of the problem. It also removes any ambiguity in the choice of a density function: any function assigning a natural density to $\mathcal{D}_t$ for every $t > 0$ agrees with $d(t)$ for every $t > 0$.

Lemma 2.2 (Divisibility monotonicity)**.** *If $n, m \geq 1$, $n \mid m$, and $t \in \Sigma(n)$, then $t \in \Sigma(m)$.*

Proof. Every divisor of n is a divisor of m. A representing subset of $\mathcal{D}(n)$ remains a representing subset of $\mathcal{D}(m)$. □

2.1. **The finite prime model and actual natural density.** Let P be a finite set of primes. On its subsets define

$$\mathbb{P}_P(T) = \prod_{p\in T} \frac{1}{p} \prod_{p\in P\setminus T} \left(1 - \frac{1}{p}\right), \qquad T \subseteq P. \tag{2.2}$$

This is a probability measure, and it is the exact natural-density law of the set of primes in P that divide an integer. Indeed, the relevant event is periodic modulo $\prod_{p\in P} p$, and the Chinese remainder theorem gives its proportion as (2.2). Write $n_T = \prod_{p\in T} p$, with $n_\varnothing = 1$.

Proposition 2.3 (Finite witnesses give density lower bounds)**.** *For every $t \geq 0$ and every finite prime set P,*

$$\sum_{T\subseteq P} \mathbb{P}_P(T)\mathbf{1}_{\{t\in\Sigma(n_T)\}} \leq d(t). \tag{2.3}$$

More generally, if $\mathcal{H}$ is a nonempty finite set of positive targets and $w_t > 0$ for $t \in \mathcal{H}$, define

$$F(T) = \frac{\sum_{t\in\mathcal{H}} w_t \mathbf{1}_{\{t\in\Sigma(n_T)\}}}{\sum_{t\in\mathcal{H}} w_t}. \tag{2.4}$$

Then $0 \leq F(T) \leq 1$ and

$$\mathbb{E}_P F \leq \frac{\sum_{t\in\mathcal{H}} w_t d(t)}{\sum_{t\in\mathcal{H}} w_t}. \tag{2.5}$$

Proof. If an integer has prime-divisibility pattern T on P, then n_T divides that integer. By Lemma 2.2, a representation using divisors of n_T is also a representation using divisors of the integer. The disjoint union of the successful patterns is therefore contained in $\mathcal{D}_t$. Taking natural densities gives (2.3). Multiplying this inequality by w_t, summing over $\mathcal{H}$, and interchanging two finite sums proves (2.5). The bounds on F follow directly from its definition. □

In the lower construction we take $w_t = 1/t$. The observable is thus the fraction of harmonic target mass represented by the selected prime product. Its connection to $d(t)$ is the exact inequality (2.5).

3. The upper bound

Put

$$\delta = 1 - \frac{1 + \log\log 2}{\log 2}, \qquad \alpha = 2 - \delta = 1 + \frac{1 + \log\log 2}{\log 2}.$$

Thus $0 < \delta < 1$ and $\alpha > 0$. We prove the following statement for the density $d(t)$ constructed in Section 2.

Theorem 3.1. *As $t \to \infty$ through the positive integers,*

$$(\log t)^{\delta} d(t) \longrightarrow 0.$$

The proof uses the divisor-interval method underlying [2]. We give the finite estimates needed here, including the ordered-volume estimate. In particular, the proof does not take a divisor-interval asymptotic as an input. The saving obtained below for a fixed-ratio divisor interval is $(\log\log y)^{-5/4}$. The final dyadic cover uses $O(\log\log t)$ intervals; retaining a saving of more than one power at the preceding stage is therefore essential.

3.1. **Finite prime laws and logarithmic support.** For a finite set P of primes, let $T \subseteq P$ be chosen by including each $p \in P$ independently with probability $1/p$. Its exact mass is

$$\mu_P(T) = \prod_{p \in T} \frac{1}{p} \prod_{p \in P \setminus T} \left(1 - \frac{1}{p}\right).$$

Write

$$w_p = \frac{1}{p-1}, \qquad w(S) = \prod_{p \in S} w_p, \qquad a(S) = \prod_{p \in S} p.$$

Empty products are 1. For a real $d \geq 0$, define the compact set

$$\mathcal{L}_d(S) = \bigcup_{U \subseteq S} \left[\sum_{p \in U} \log p,\ \sum_{p \in U} \log p + d \right], \qquad L_d(S) = |\mathcal{L}_d(S)|,$$

where $|\cdot|$ denotes Lebesgue measure. The unnormalized support sum is

$$Z_d(P) = \sum_{S \subseteq P} w(S) L_d(S).$$

For every $I \subseteq S$,

$$L_d(S) \leq 2^{|S|-|I|} \left(d + \sum_{p \in I} \log p \right), \qquad L_d(S \cup \{p\}) \leq 2 L_d(S). \tag{3.1}$$

Indeed, fixing the chosen subset of $S \setminus I$ leaves one translate of a set contained in $[0, d + \sum_{p \in I} \log p]$. The second inequality follows by separating subsets according to whether they contain p. These inequalities hold even when some of the translated intervals overlap.

We use the elementary Mertens estimates in the following uniform form: there are absolute constants $m \in \mathbb{R}$ and $E > 0$ such that, for every real $x \geq 2$,

$$\left| \sum_{p \leq x} \frac{1}{p} - \log\log x - m \right| \leq \frac{E}{\log x}, \qquad \sum_{p \leq x} \frac{\log p}{p} \leq \log x + \log 4 + 4. \tag{3.2}$$

All sums over p in this section are over primes. These are the Mertens estimates proved in the imported library and used by the formal argument; no uniform prime number theorem in short intervals is used.

Lemma 3.2 (A support-weighted logarithmic moment)**.** *There is an absolute constant $K_4 > 0$ such that, for every real $x \geq 2$, every finite prime set $P \subseteq [2, x]$, and every $d \geq 0$,*

$$\sum_{S \subseteq P} w(S) L_d(S) \left(\sum_{p \in S} \log p \right)^4 \leq K_4 (\log x)^4 Z_d(P).$$

Proof. More generally, suppose $F(S) \geq 0$ and $F(S \cup \{p\}) \leq 2F(S)$. For numbers $0 \leq b_p \leq 1$, expansion of $\prod_{p \in S}(1 + (e^{b_p} - 1))$ gives

$$\begin{aligned} \sum_{S \subseteq P} w(S) F(S) e^{\sum_{p \in S} b_p} &\leq \left(\sum_{S \subseteq P} w(S) F(S) \right) \prod_{p \in P} \left(1 + 2w_p (e^{b_p} - 1)\right) \\ &\leq \left(\sum_{S \subseteq P} w(S) F(S) \right) \exp\left(2(e-1) \sum_{p \in P} w_p b_p \right). \end{aligned}$$

For the first inequality, a term indexed by $U \subseteq S$ is reindexed as $S = U \cup R$, with $R \cap U = \varnothing$, and $F(U \cup R) \leq 2^{|U|} F(R)$. Dropping the restriction $R \cap U = \varnothing$ then increases the sum. The second inequality uses $e^b - 1 \leq (e-1)b$ for $0 \leq b \leq 1$.

Take $F = L_d$ and $b_p = \log p / \log x$. By (3.2) and $1/(p-1) \leq 2/p$,

$$\sum_{p \in P} w_p b_p \leq K_0 := 2 \left(1 + \frac{\log 4 + 4}{\log 2} \right).$$

Finally $z^4 \leq 24 e^z$ for $z \geq 0$. The conclusion follows with $K_4 = 24 \exp(2(e-1)K_0)$. □

3.2. **From actual primes to a continuous envelope.** For an integer $N \geq 2$, put

$$M_N = \sum_{p \leq N} \frac{1}{p-1}, \qquad V_N = \left\lceil \frac{M_N}{\log 2} \right\rceil.$$

The correction $1/(p-1) - 1/p = 1/(p(p-1))$ has total sum at most 1. Consequently (3.2) implies

$$M_N = \log\log N + O(1), \qquad V_N \log 2 = \log\log N + O(1), \tag{3.3}$$

with absolute, uniform bounds for the two errors.

Associate to each prime $p \leq N$ the half-open interval

$$J_p = (M_{p-1}, M_p].$$

These intervals are pairwise disjoint and have length w_p. There is an absolute $K_c > 0$ such that

$$\log p \leq K_c e^u \quad (u \in J_p). \tag{3.4}$$

For example, one may take $K_c = \exp(-m + E/\log 2 + 1)$: the lower estimate in (3.2) gives $\log\log p \leq M_{p-1} - m + E/\log 2 + 1$.

For $k \geq 0$ and $u = (u_1, \ldots, u_k) \in \mathbb{R}^k$, set

$$H_k(u) = \min_{I \subseteq \{1, \ldots, k\}} \frac{1 + \sum_{i \in I} e^{u_i}}{2^{|I|}}.$$

In particular, $0 \le H_k \le 1$. For $v > 0$, write

$$\mathcal{E}_k(v) = \int_{[0,1]^k} H_k((\log 2)vx_1, \dots, (\log 2)vx_k)\, dx.$$

Lemma 3.3 (Exact cell comparison)**.** *For every finite prime set $P \subseteq [2, N]$, every $d \ge 0$, every $k \ge 0$, and every real $v > 0$ with $M_N \le v \log 2$,*

$$\sum_{\substack{S \subseteq P \\ |S|=k}} w(S) L_d(S) \le (d + K_c) \frac{(2v \log 2)^k}{k!} \mathcal{E}_k(v). \tag{3.5}$$

Proof. For an injective ordered tuple of primes $(p_1, \dots, p_k)$, the box $\prod_i J_{p_i}$ has volume $\prod_i w_{p_i}$. On this box, (3.1) and (3.4) give

$$L_d(\{p_1, \dots, p_k\}) \le (d + K_c) 2^k H_k(u).$$

Boxes belonging to different tuples are disjoint. Integrate this inequality, sum over injective tuples, and enlarge their union to $[0, M_N]^k$. Each k-element prime set has exactly $k!$ orderings. Finally enlarge to $[0, v \log 2]^k$ and rescale the coordinates. This proves (3.5); repeated-prime tuples have not been assigned any arithmetic interpretation. □

3.3. **An ordered barrier estimate.** We now prove the geometric estimate used to control $\mathcal{E}_k$. For $k \ge 1$ and real u, v, let

$$G_k(u, v) = \{0 \le x_1 \le \cdots \le x_k \le 1 : i - u \le v x_i \ (1 \le i \le k)\}.$$

Lemma 3.4 (Ordered barrier)**.** *If $k \ge 1$, $u, w \ge 0$, and $v = k + w - u$, then*

$$k!\, |G_k(u, v)| \le \frac{36(u + 1)(w + 1)^2}{k}. \tag{3.6}$$

Proof. An ordered j-simplex in an interval of length $a \ge 0$ has volume $a^j/j!$, including $j = 0$. This follows by partitioning the cube into its $j!$ permutation chambers; the coordinate-tie hyperplanes have measure zero.

First suppose $v \ge k/2$ and $2(2w + 2) \le v$. Put $c = 2w + 2$ and $T = 1 - c/v$. Partition the bad part of the ordered simplex with endpoint T by its first failed inequality. If the first failure is at coordinate $\ell + 1$, put $m = k - \ell - 1$ and call that coordinate t. Fubini gives its exact volume as

$$\int_{\substack{0 \le t \le T \\ vt < \ell + 1 - u}} |G_\ell(t; u, v)| \, \frac{(T - t)^m}{m!}\, dt, \tag{3.7}$$

where $G_\ell(t; u, v)$ denotes the ordered prefix in $[0, t]$ satisfying the same first ℓ inequalities. The prefix factor is important: when it is nonzero, $\ell - u \le vt$. Thus

$$\lambda = \ell + 1 - u - vt \in (0, 1], \qquad A = w + 2 - \lambda, \quad B = w + \lambda$$

satisfy $0 \le B \le A \le m$. On the domain of (3.7), $m > 0$ and

$$\frac{T - t}{1 - t} = \frac{m - A}{m + B}, \qquad A + B = c.$$

The elementary inequality

$$\left(\frac{m - A}{m + B}\right)^m \le e^{-A-B} \quad (0 \le B \le A \le m)$$

therefore shows that each first-crossing volume at endpoint T is at most e^{-c} times the corresponding volume at endpoint 1. For completeness, if $A < m$, take logarithms and use $\log(1-a) - \log(1+b) \le -a-b$ for $0 \le b \le a < 1$; the case $A = m$ has zero left side. The logarithmic inequality follows from $\log(1 - a) \le -2a/(2 - a)$ and $\log(1 + b) \ge 2b/(2 + b)$.

There is no good tuple at endpoint T: its last inequality would require $k-u \le vT = k-u-w-2$. Summing the disjoint first-crossing cells gives

$$\frac{T^k}{k!} \le e^{-c}\left(\frac{1}{k!} - |G_k(u,v)|\right).$$

Hence

$$k!|G_k(u,v)| \le 1 - e^c(1-c/v)^k. \tag{3.8}$$

Because $0 \le c/v \le 1/2$, the inequality $\log(1-z) \ge -z-2z^2$ gives

$$c + k\log(1-c/v) \ge -\frac{4u(w+1)+32(w+1)^2}{k}.$$

Indeed, the linear term is $2(w+1)(w-u)/v$, bounded below by $-4u(w+1)/k$, and the quadratic term is bounded below by $-32(w+1)^2/k$. Applying $e^r \ge 1+r$ in (3.8) proves (3.6).

In the remaining regimes use $k!|G_k(u,v)| \le 1$. If $v < k/2$, the relation $v = k+w-u$ implies $u > k/2$. If $v \ge k/2$ but $v < 4w+4$, then $k < u+3w+4 \le 4(u+1)(w+1)^2$. In either case the right side of (3.6) is at least 1. □

For $x \in [0,1]^k$, let $x_{(1)} \le \cdots \le x_{(k)}$ be its sorted coordinates and define the nonpositive deficit

$$D_v(x) = \min\bigl(0, \min_{1\le i\le k}(vx_{(i)} - i)\bigr).$$

The permutation-chamber decomposition and Lemma 3.4 imply, whenever $s \ge 0$ and $s+v-k \ge 0$,

$$\mathbb{P}\{D_v \ge -s\} \le \frac{36(s+1)(s+v-k+1)^2}{k}, \tag{3.9}$$

where probability is normalized Lebesgue measure on the unit cube. For $v \ge 0$ one also has $D_v \le -\max(0, k-v)$.

Lemma 3.5 (Deficit moment)**.** *For every integer $k \ge 1$ and every real $v \ge 0$,*

$$\mathbb{E}\,2^{D_v} \le \frac{27648(1+(k-v)^2)}{k(1+2^{k-v})}. \tag{3.10}$$

Proof. Put $b = k-v$ and $r = \max(0,b)$. Decomposing the negative deficit into unit layers gives the pointwise bound

$$2^{D_v} \le \sum_{j\ge 0} 2^{-r-j}\mathbf{1}_{\{D_v \ge -r-j-1\}}.$$

By (3.9), the probability in the jth term is at most $72(1+b^2)(j+2)^3/k$. Here we used

$$(r+j+2)(r-b+j+2)^2 \le 2(1+b^2)(j+2)^3.$$

Since $\sum_{j\ge 0} 2^{-j}(j+2)^3 \le 192$, integration gives $13824(1+b^2)2^{-r}/k$. Finally $2^{-\max(0,b)} \le 2/(1+2^b)$. □

3.4. **Occupancy and the full support sum.**

Lemma 3.6 (Envelope moment)**.** *There is an integer v_0 such that, for every integer $v \ge v_0$ and every integer $1 \le k \le 10v$,*

$$\mathcal{E}_k(v) \le \frac{500000\, v^{1/4}(1+(k-v)^2)}{(k+1)(1+2^{k-v})}. \tag{3.11}$$

Proof. Partition $[0,1)$ into v equal bins and let $R \ge 1$ be an integer. The probability that some bin contains at least R coordinates is at most

$$v\binom{k}{R}v^{-R}. \tag{3.12}$$

This is the union bound over the bin and the R coordinate indices. Coordinates on bin endpoints cause no change in probability.

On the complementary event every bin contains at most R coordinates. For a sorted prefix ending at $x_{(i)}$, summation over bins gives

$$\sum_{j\le i} 2^{\nu x_{(j)}} \le 4R\, 2^{\nu x_{(i)}}.$$

Using this prefix in the definition of H_k, as well as the empty subset, shows

$$H_k((\log 2)\nu x) \le (4R+1)2^{D_\nu(x)}.$$

On the exceptional event the full subset and the empty subset give the bound $\min(1,(k+1)2^{\nu-k})$. Thus the following finite inequality holds before R is chosen:

$$\begin{aligned}\mathcal{E}_k(\nu) &\le (4R+1)\frac{27648(1+(k-\nu)^2)}{k(1+2^{k-\nu})}\\ &\quad+\min(1,(k+1)2^{\nu-k})\,\nu\binom{k}{R}\nu^{-R}.\end{aligned}\tag{3.13}$$

Take $R=\lceil \nu^{1/4}\rceil$. Uniformly for $k\le 10\nu$,

$$\nu\binom{k}{R}\nu^{-R} \le \frac{\nu 10^R}{R!} \le \nu e^{20}2^{-R} \le \nu^{-3}$$

for all sufficiently large ν; the middle inequality follows from $20^R/R!\le e^{20}$. Also $R\le 2\nu^{1/4}$ and

$$\min(1,(k+1)2^{\nu-k}) \le \frac{2(k+1)}{1+2^{k-\nu}}.$$

Substitution in (3.13), using $k+1\le 2k$, $k+1\le 11\nu$, and $\nu^{1/4}\ge 1$, proves (3.11). The displayed constant is a convenient uniform choice; its value has no asymptotic significance. □

Proposition 3.7 (Support sum)**.** *There are absolute constants $K_s>0$ and $N_0\in\mathbb{N}$ such that, for every integer $N\ge N_0$, every finite prime set $P\subseteq[2,N]$, and every real $d\ge 0$,*

$$Z_d(P) \le 2d + K_s(d+K_c)\frac{(\log N)^\alpha}{(\log\log N)^{5/4}}.\tag{3.14}$$

Proof. Take $\nu=V_N$. Set

$$A_{\nu,k}=\frac{(2\nu\log 2)^k}{(k+1)!},\qquad B_{\nu,k}=A_{\nu,k}\frac{1+(k-\nu)^2}{1+2^{k-\nu}}.$$

For $\nu\ge 10$, the factorial ratios give

$$A_{\nu,k}\le (4/5)^{\nu-k}A_{\nu,\nu}\quad(k\le\nu),\qquad A_{\nu,\nu+j}\le(8/5)^jA_{\nu,\nu}\quad(j\ge 0).$$

The factor $1+2^{k-\nu}$ therefore yields, on either side of ν,

$$B_{\nu,k}\le A_{\nu,\nu}(4/5)^{|k-\nu|}(1+|k-\nu|^2).$$

In particular, the sum of $B_{\nu,k}$ over any finite set of nonnegative integers is at most $500A_{\nu,\nu}$. Lemmas 3.3 and 3.6 now bound the contribution of $1\le|S|\le 10\nu$ by

$$250000000(d+K_c)\nu^{1/4}A_{\nu,\nu}.\tag{3.15}$$

The empty set contributes d. For the remaining tail, use $L_d(S)\le d2^{|S|}$ and the exact generating function

$$\sum_{S\subseteq P} w(S)4^{|S|}=\prod_{p\in P}(1+4w_p)\le\exp(4M_N)\le 2^{4\nu}.$$

On $|S|>10\nu$, $2^{|S|}\le 2^{-10\nu}4^{|S|}$. The tail is thus at most d, as asserted in (3.14).

Stirling's lower bound, including its square-root factor, gives

$$A_{\nu,\nu}\le\frac{\exp((\log 2+1+\log\log 2)\nu)}{\nu^{3/2}}.$$

Multiplication by $v^{1/4}$ in (3.15) leaves $v^{-5/4}$. Finally (3.3) converts the exponential numerator to an absolute constant times $(\log N)^{\alpha}$ and gives $v \gg \log\log N$. This proves the proposition with one absolute N_0, uniformly in P and d. □

3.5. **Stopping at the largest prime of a witnessing divisor.** For a real $y \geq 0$, let $\mathcal{R}_y(T)$ be the event

$$\exists U \subseteq T : \quad y < a(U) \leq 2y.$$

Write $R_P(y) = \sum_{T \subseteq P} \mu_P(T) \mathbf{1}_{\mathcal{R}_y(T)}$ and $R(y) = R_{\{p : p \leq 2y\}}(y)$. A prime larger than $2y$ cannot belong to a witnessing subset; projection of the finite product law and monotonicity therefore give

$$R_P(y) \leq R(y) \tag{3.16}$$

for every finite prime set P and every $y \geq 0$.

We first record the support-to-prime-mass estimate, with endpoints included. If $\log 2 \leq a \leq b$, then (3.2) implies

$$\sum_{a \leq \log p \leq b} \frac{1}{p} \leq \frac{b - a + 2E + 1}{a}. \tag{3.17}$$

For the open-left interval subtract the two Mertens estimates and use $\log(b/a) \leq (b-a)/a$. There is at most one prime at the left endpoint; its mass is $e^{-a} \leq 1/a$, explaining the final 1.

Lemma 3.8 (Packing support intervals)**.** *Let $A \subset \mathbb{R}$ be finite, $d > 0$, and $U = \bigcup_{a \in A} [a, a+d]$. For every $v \geq \log 2$,*

$$\sum_{\substack{p \text{ prime} \\ \log p \geq v,\ \log p \in U}} \frac{1}{p} \leq \frac{3d + 2E + 1}{d} \frac{|U|}{v}.$$

Proof. Choose a maximal subset $J \subseteq A$ whose distinct points have distance greater than d. The intervals $[b, b+d]$, $b \in J$, are disjoint, so $|J|d \leq |U|$. Maximality shows that every $a \in A$ is within d of a point of J, whence $U \subseteq \bigcup_{b \in J} [b - d, b + 2d]$. Intersect each covering interval with $[v, \infty)$ and discard empty intersections. Each surviving interval has length at most $3d$ and lower endpoint at least v. Apply (3.17), sum over J, and use $|J|d \leq |U|$. □

For a finite prime set S, let $p_*(S) = \max S$, with $p_*(\varnothing) = 0$, and define

$$D_y(S) = \log\left(p_*(S) + \frac{y}{a(S)}\right).$$

If $y \geq 2$, this is positive and at least $\log 2$.

Lemma 3.9 (Stopping-prime localization)**.** *There is an absolute constant $K_\ell > 0$ with the following property. If $y \geq 2$ and P is a finite prime set containing every prime smaller than each of its elements, then*

$$R_P(y) \leq K_\ell \sum_{S \subseteq P} \frac{w(S) L_{\log 2}(S)}{D_y(S)^2}. \tag{3.18}$$

Proof. Choose the largest prime p of a witnessing subset $U \subseteq T$. Since $y > 1$, the witness is nonempty. Put $P_{<p} = \{q \in P : q < p\}$ and $S = T \cap P_{<p}$. Then some $V \subseteq S$ satisfies $y < pa(V) \leq 2y$. The exact probability that $p \in T$ and $T \cap P_{<p} = S$, with no restriction above p, is

$$\frac{1}{p} \prod_{q \in P_{<p}} (1 - 1/q)\, w(S).$$

The union bound over p, S thus bounds $R_P(y)$ by the sum of these quantities over stopping witnesses. Reindex this finite sum by S.

For each eligible p one has $p \geq p_*(S)$ and $p > y/a(S)$; consequently $D_y(S) \leq \log(2p) \leq 2\log p$. Moreover $\log p$ belongs to

$$U_y(S) = \bigcup_{V \subseteq S} [\log y - \log a(V),\ \log y - \log a(V) + \log 2].$$

Reflection preserves measure and gives $|U_y(S)| = L_{\log 2}(S)$. Set $v = \max(\log 2, D_y(S)/2)$. Every eligible prime has $\log p \geq v$. The completeness hypothesis on P and (3.2) imply

$$\prod_{q \in P_{<p}} (1 - 1/q) \leq \exp\left(-\sum_{q<p} 1/q\right) \leq \frac{K_c}{\log p} \leq \frac{K_c}{v}.$$

Lemma 3.8, applied to $U_y(S)$ with $d = \log 2$, supplies a second factor $1/v$. Since $v \geq D_y(S)/2$, the inner sum over p is at most an absolute constant times $L_{\log 2}(S)/D_y(S)^2$. Multiplication by $w(S)$ and summation prove (3.18). □

Lemma 3.10 (Removing the denominator)**.** *There is an absolute $K_D > 0$ such that, for every $x \geq 2$, every finite prime set $P \subseteq [2, x]$, and every $d \geq 0$,*

$$\sum_{S \subseteq P} \frac{w(S) L_d(S)}{D_x(S)^2} \leq \frac{K_D}{(\log x)^2} Z_d(P). \tag{3.19}$$

Proof. Write $L = \log x$. The definition gives

$$\log p \leq D_x(S) \quad (p \in S), \qquad L \leq D_x(S) + \sum_{p \in S} \log p.$$

The sets with $D_x(S) \geq L/4$ contribute at most $16 Z_d(P)/L^2$. The empty set is in this class. For a remaining set write $S = \{p\} \cup R$, where p is its largest prime. Then

$$\sum_{q \in R} \log q \geq L/2, \qquad \frac{L_d(S)}{D_x(S)^2} \leq \frac{2 L_d(R)}{(\log p)^2}.$$

Apply Lemma 3.2 in the prime universe $P_{<p}$, with cutoff p, and use the fourth-moment bound on the event $\sum_{q \in R} \log q \geq L/2$. For this fixed largest prime the inner sum is at most

$$\frac{32 K_4 (\log p)^2}{L^4} Z_d(P_{<p}) \leq \frac{32 K_4 (\log p)^2}{L^4} Z_d(P).$$

After multiplication by w_p and summation over p, use

$$\sum_{p \in P} w_p (\log p)^2 \leq L \sum_{p \in P} w_p \log p \leq K_0 L^2.$$

Thus $K_D = 16 + 32 K_4 K_0$ suffices. □

Since $D_{2y}(S) \leq \log 2 + D_y(S) \leq 2 D_y(S)$ for $y \geq 2$, the last two lemmas, applied to $P = \{p : p \leq 2y\}$, give

$$R(y) \leq \frac{K_R}{(\log(2y))^2} Z_{\log 2}(\{p : p \leq 2y\}) \tag{3.20}$$

with an absolute $K_R > 0$. Taking $N = \lfloor 2y \rfloor$ in Proposition 3.7, and enlarging constants to absorb this floor, proves that absolute $K_1, K_2 > 0$ and y_0 exist such that, for every real $y \geq y_0$,

$$R(y) \leq \frac{K_1}{(\log(2y))^2} + \frac{K_2}{(\log(2y))^{\delta} (\log\log(2y))^{5/4}}. \tag{3.21}$$

This is an estimate for the actual finite Bernoulli law, not for a continuous surrogate law.

3.6. **Restoring prime powers.** Call a positive integer B powerful if every prime dividing B divides it to exponent at least 2; include $B = 1$. We need the two absolute bounds

$$\sum_{B \text{ powerful}} \frac{\tau(B)}{B} < \infty, \qquad \sum_{\substack{B \text{ powerful} \\ B>L}} \frac{1}{B} \le K_p L^{-1/4} \quad (L > 0). \tag{3.22}$$

Here $\tau(B)$ is the number of positive divisors. To verify them, write each powerful integer uniquely as a^2b^3, with b squarefree, by separating the parity of each prime exponent. Thus $\sum_{B \text{ powerful}} B^{-3/4}$ converges by comparison with $\sum_{a,b\ge1} a^{-3/2}b^{-9/4}$. The elementary bound $\tau(n) \le K_\tau n^{1/4}$ now proves the first assertion. One proof of this bound uses $(e+1) \le p^{e/4}$ for $p \ge 16$, and absorbs the finite suprema of $(e+1)p^{-e/4}$ for primes $p < 16$ into K_τ. The formal proof supplies the explicit choice $K_\tau = 768^4$. For the second assertion, use $B^{-1} \le L^{-1/4}B^{-3/4}$ when $B > L$.

Let $h(y, z)$ denote the natural density of integers with a divisor in $(y, z]$, for integers $y, z \ge 0$ (the interval is empty if $y \ge z$). This density exists because the condition is periodic. The following finite construction accounts for all prime powers in this density.

Fix a finite prime set P, and integers $c_p \ge 2$ for $p \in P$. For $0 \le e_p \le c_p$, give the vector $e = (e_p)_{p\in P}$ the product weight with single-prime factors

$$\nu_{p,c_p}(e_p) = \begin{cases} (1 - 1/p)p^{-e_p}, & e_p < c_p, \\ p^{-c_p}, & e_p = c_p. \end{cases} \tag{3.23}$$

The last atom records all valuations at least c_p. The Chinese remainder theorem identifies this finite law with the density of capped valuation patterns. If P contains every prime at most z and $c_p \ge z$, whether an integer has a divisor in $(y, z]$ is determined by the finite product $\prod_{p\in P} p^{e_p}$. Hence $h(y, z)$ is exactly the sum of (3.23) over the appropriate finite event.

Lemma 3.11 (Powerful-part domination)**.** *For any capped law* (3.23)*, any real $y \ge 0$, and any real $L > 0$, its probability of a divisor in $(y, 2y]$ is at most*

$$\sum_{\substack{B\in\mathcal{B} \\ B\le L}} \frac{1}{B} \sum_{f|B} R(y/f) + \sum_{\substack{B\in\mathcal{B} \\ B>L}} \frac{1}{B}, \tag{3.24}$$

where $\mathcal{B}$ is the finite set of powerful parts of its exponent vectors.

Proof. For an exponent vector put

$$H = \{p : e_p \ge 2\}, \quad B = \prod_{p\in H} p^{e_p}, \quad T = \{p : e_p = 1\}.$$

Then the corresponding integer is $Ba(T)$, and $\gcd(B, a(T)) = 1$. Every divisor factors uniquely as $fa(U)$ with $f \mid B$ and $U \subseteq T$. Thus a divisor in $(y, 2y]$ implies $\mathcal{R}_{y/f}(T)$ for at least one $f \mid B$.

Fix H and its exponents. Unique factorization makes this the same as fixing B; no different high-exponent patterns produce the same integer B. For $p \notin H$, the total mass of exponents $0, 1$ is $1 - p^{-2}$, and their normalized probability of exponent 1 is exactly $1/(p + 1)$. In particular, the exact mass on this fiber is

$$m_B = \prod_{p\in H} \nu_{p,c_p}(e_p) \prod_{p\in P\setminus H} (1 - p^{-2}) \le \frac{1}{B},$$

times the independent subset law on $P \setminus H$ with selection probabilities $1/(p+1)$. This factorization also holds when some $e_p = c_p$, because the terminal atom in (3.23) has its stated mass p^{-c_p}.

The event $\mathcal{R}_z(T)$ is increasing in T. The conditional law is therefore dominated by independent selection with probabilities $1/p$ on P, and (3.16) bounds each event by $R(z)$. For $B \le L$ apply the union bound over $f \mid B$ and multiply by $m_B \le 1/B$. For $B > L$ use the whole fiber mass, rather than the divisor union bound. Finally sum over the finite set $\mathcal{B}$. □

Taking $L = \sqrt{y}$ in (3.24) retains the logarithmic saving. Indeed, if $f \mid B \le \sqrt{y}$, then $\sqrt{y} \le y/f \le y$. Uniformly in this range the right side of (3.21) is at most

$$\frac{K_1'}{(\log y)^2} + \frac{K_2'}{(\log y)^{\delta}(\log\log y)^{5/4}}$$

for all sufficiently large y. Apply the first bound in (3.22) to the small-B contribution and the second to the remaining fibers. Choosing the capped law with all primes at most $2y$ and caps $2y + 2$ gives the following consequence.

Proposition 3.12 (A uniform divisor-interval envelope)**.** *There are absolute positive constants A_1, A_2, A_3 and an integer y_1 such that, for every integer $y \ge y_1$,*

$$h(y, 2y) \le F(y) := \frac{A_1}{(\log y)^2} + \frac{A_2}{(\log y)^{\delta}(\log\log y)^{5/4}} + \frac{A_3}{y^{1/8}}. \tag{3.25}$$

Moreover, F is nonnegative and decreasing for all sufficiently large real arguments, and

$$(\log y)^{\delta} \log\log y\, F(y) \longrightarrow 0. \tag{3.26}$$

Proof. The preceding finite conditioning and summability estimates establish (3.25) with absolute constants and one threshold valid for all y. The term $y^{-1/8}$ is $(\sqrt{y})^{-1/4}$. Positivity and eventual monotonicity follow termwise from $\delta > 0$. After multiplication by $(\log y)^{\delta} \log\log y$, the three terms are respectively

$$A_1 \frac{\log\log y}{(\log y)^{\alpha}}, \qquad A_2 (\log\log y)^{-1/4}, \qquad A_3 \frac{(\log y)^{\delta} \log\log y}{y^{1/8}}.$$

All tend to zero. These are precisely the three limiting comparisons used in the formal upper-bound argument. □

3.7. **The dyadic reduction for divisor sums.**

Proof of Theorem 3.1. For positive integers t and integers $a \ge 0$,

$$d(t) \le h(a, t) + \frac{a}{t}. \tag{3.27}$$

To see this, if t is a sum of distinct divisors of n and n has no divisor in $(a, t]$, the sum of all divisors of n at most a is at least t. Its mean over integers n is a: each divisor j contributes $j \cdot (1/j) = 1$. Averaging the resulting pointwise indicator inequality proves (3.27).

For an integer $J \ge 0$ put $a_j = \lfloor t/2^j \rfloor$. The intervals $(a_{j+1}, a_j]$, $0 \le j < J$, cover $(a_J, t]$ exactly. Also $a_j \le 2a_{j+1} + 1$, so each interval contributes at most $h(a_{j+1}, 2a_{j+1}) + 1/(2a_{j+1} + 1)$; the last term is the density of multiples of the possible extra endpoint. Finite subadditivity and (3.27) therefore give the exact bound

$$d(t) \le \sum_{j=0}^{J-1} h(a_{j+1}, 2a_{j+1}) + \frac{J}{2a_J + 1} + 2^{-J}. \tag{3.28}$$

Choose

$$J = \left\lceil \frac{\log\log t}{\log 2} \right\rceil.$$

For all sufficiently large t,

$$\log t \le 2^J < 2\log t, \qquad J \le (1 + 1/\log 2)\log\log t, \qquad a_J \ge \sqrt{t}.$$

The last assertion follows from $4\log t \le \sqrt{t}$ and the elementary floor inequalities, and includes the rounding in a_J. All the divisor intervals in (3.28) therefore have lower endpoint at least $\sqrt{t}$. Proposition 3.12 and eventual monotonicity give a common upper bound $F(\sqrt{t})$. After multiplication by $(\log t)^{\delta}$, the first term is at most

$$(1 + 1/\log 2)(\log t)^{\delta} \log\log t\, F(\sqrt{t}),$$

which tends to zero by (3.26) and $\log\sqrt{t} = \frac{1}{2}\log t$. The two error terms in (3.28) are bounded, after the same multiplication, by a constant times $(\log t)^{\delta+1}/\sqrt{t}$ and by $(\log t)^{\delta-1}$. They tend to zero because $\delta < 1$. Since $d(t) \geq 0$, the theorem follows. □

4. The lower logarithmic rate

Write $d(t)$ for the density of integers for which t is a sum of distinct positive divisors. We prove the lower-density assertion in the following form. As throughout the paper, logarithms are natural.

Theorem 4.1. *Let*

$$\lambda_* = (\log 2)^{-1}, \qquad \delta = 1 - \frac{1 + \log\log 2}{\log 2}.$$

For every $\varepsilon > 0$ there are arbitrarily large integers t such that

$$d(t) \geq \exp\bigl(-(\delta+\varepsilon)\log\log t\bigr).$$

Consequently $\liminf_{t\to\infty} -\log d(t)/\log\log t \leq \delta$.

The proof constructs a finite prime configuration with three properties. It contains a divisor system with many independent bounded integer combinations; its divisor logarithms have extremely small uncovered measure on a long interval; and an additional collection of large primes eliminates nonzero rational phases for that divisor system. A small practical factor then converts bounded integer coefficients into distinct divisors. The probability estimate is made in a tilted finite product model, and is transferred to natural density only after averaging over targets.

4.1. **The exact finite model and its sampling identities.** For a finite set P of primes and $T \subseteq P$, put

$$N(T) = \prod_{p\in T} p, \qquad w_P(T) = \prod_{p\in T}\frac{1}{p}\prod_{p\in P\setminus T}\left(1-\frac{1}{p}\right).$$

This is the density of the corresponding divisibility pattern. For $\lambda > 0$, define

$$Z_P(\lambda) = \prod_{p\in P}\left(1+\frac{\lambda-1}{p}\right), \qquad \mathbb{P}_{P,\lambda}(T) = \frac{\lambda^{|T|}w_P(T)}{Z_P(\lambda)}. \tag{4.1}$$

The coordinates remain independent, with inclusion probability $z_p = \lambda/(p+\lambda-1)$. We will use the finite Harris inequality: increasing events are positively correlated under this law. It follows by induction on $|P|$ from the two-point covariance identity.

Two distinct auxiliary sampling laws occur below. Set

$$a_p = \log\left(1+\frac{\lambda}{p-1}\right), \qquad \mu_Q = \sum_{p\in Q} a_p$$

for a prime window Q. The identity

$$\mathbb{P}_{Q,\lambda}(S) = e^{-\mu_Q}\prod_{p\in S}\frac{\lambda}{p-1} \tag{4.2}$$

has two useful consequences. First, conditioned on $|S| = m$, the actual subset law has weights proportional to $\prod_{p\in S}(p-1)^{-1}$. Second, if $p_1,\dots,p_m$ are independent with probabilities a_p/μ_Q, and E is a property of their underlying set, then

$$\mathbb{P}_{Q,\lambda}(|S| = m,\ E(S)) \geq e^{-\mu_Q}\frac{\mu_Q^m}{m!}\left(\mathbb{P}(E(\{p_1,\dots,p_m\})) - m^2\max_{p\in Q}\frac{a_p}{\mu_Q}\right). \tag{4.3}$$

Indeed, remove tuples having a repeated coordinate, use $\lambda/(p-1) \geq a_p$, and divide by the $m!$ orderings of each distinct set. The subtracted term is the union bound over repeated pairs. No limiting Poisson process or Poisson approximation is used in this identity. In particular, conditioning the actual subset count is never identified with independent sampling from a_p/μ_Q.

We use a quantitative reciprocal-prime estimate in the form

$$\sum_{u<\log p\le v}\frac{1}{p}=\log(v/u)+O(1/u)\qquad(\log 2\le u\le v). \tag{4.4}$$

There is an absolute constant $C_0>0$ for which, uniformly for $1\le\lambda\le 2$,

$$\left|\lambda^{-1}\sum_{u<\log p\le v}a_p-\log(v/u)\right|\le\frac{2C_0}{u}. \tag{4.5}$$

This follows from (4.4), the elementary inequalities for $\log(1+x)$, and the convergent sum of reciprocal prime squares. We increase C_0 when necessary. All prime windows below have the form $\{p : u < \log p \le v\}$, with this endpoint convention.

We also record the finite conditional-independence rule used for the final adaptive window. If $A\subseteq P$, and $Q(S)\subseteq P\setminus A$ is chosen from an observed $S\subseteq A$, then, for every function F,

$$\begin{aligned}&\mathbb{E}_{P,\lambda}F(T\cap A,T\cap Q(T\cap A))\\&\quad=\sum_{S\subseteq A}\mathbb{P}_{A,\lambda}(S)\sum_{U\subseteq Q(S)}\mathbb{P}_{Q(S),\lambda}(U)F(S,U).\end{aligned} \tag{4.6}$$

For each fixed S, this is just the product law on the complement of A; summing the coordinates outside $A\cup Q(S)$ gives one. Uniform conditional estimates may therefore be applied after a core and all earlier representatives have been selected.

4.2. **A proper divisor core at the ballot rate.** Fix $\lambda_*<\lambda<2$ and put

$$J(\lambda)=\lambda\log 2-1-\log(\lambda\log 2).$$

Our parameters are

$$\begin{gathered}h=2^s,\quad K=2s+4,\quad n=h-K,\quad b=2^{4h},\quad k=bh,\\A=10k^3,\quad B=\frac{501}{500}(k^2+3),\quad X=A+B2^k,\quad \eta=\frac{1}{1000},\\H=\lceil e^{k^2}\rceil,\quad Y=\lceil e^{k^3}\rceil,\quad m=\lceil k^{3/4}\rceil,\quad Z=\lceil e^{3k^3}\rceil.\end{gathered} \tag{4.7}$$

All assertions involving $n>0$ are for sufficiently large s. The quantities H and Y have different roles: H bounds coefficients in a proper cube, whereas Y^{-1} is a phase tolerance.

Partition the prime window $(A+B,X]$ in logarithmic coordinates into the k bands

$$P_j=\{p : A+B2^j<\log p\le A+B2^{j+1}\},\qquad 0\le j<k.$$

Group these bands into b consecutive blocks of length $h=n+K$. In a block starting at ℓ, require the following:

(1) The first n bands contain exactly n selected primes, and the first j of these bands contain at most j selected primes for every $0\le j\le n$.
(2) Each of the last K bands contains exactly one selected prime, and the prime in band $\ell+n+j$ satisfies

$$\log p\le A+Be^{\eta}2^{\ell+n+j}.$$

The lower endpoint of each band is unchanged in the second condition. The restriction to an early portion of the final K bands supplies the sharp endpoint bound needed for the Fourier anchors.

Lemma 4.2 (Block probability and endpoint bounds)**.** *The event just described, over all b blocks, has tilted probability*

$$\exp\bigl(-(J(\lambda)+o(1))k\bigr).$$

Here and below this notation denotes a lower bound with an $o(1)$ error in the exponent. Every selected set $\mathcal{K}$ in this event has k elements. For the prefix $\mathcal{K}_t$ consisting of the first t selected primes, at a completed block endpoint $t = ih$, one has

$$\sum_{p\in\mathcal{K}_t} \log p \le tA + \frac{B(e^{\eta} + n2^{-K})2^t}{1-2^{-h}}. \tag{4.8}$$

Moreover, if the selected primes are increasingly ordered as $p_1 < \cdots < p_k$, then

$$\log p_{j+1} \ge A + B2^j \qquad (0 \le j < k). \tag{4.9}$$

Proof. We give the probability estimate including the effects of the shifted bands. Let B' be the scale at the start of one block, and put $\xi(B') = 2C_0/(B'\log 2)$. For the first n bands, their intensity masses have increasing lower bounds

$$\lambda(1-\xi(B'))\log\frac{A+B'2^{j+1}}{A+B'2^j} \ge \frac{\lambda(1-\xi(B'))\log 2}{1+(A/B')2^{-j}}.$$

Among the n^n assignments of n labelled marks to n bins, at least n^{n-1} obey the upper-prefix ballot condition. To see this, rotate the occupancy sequence: a rotation starting just after a maximum of the partial sums of "occupancy minus one" has all partial sums nonpositive. Every assignment is covered by at most n rotations. If an assignment is ballot, its ith smallest bin index is at least i (indices start at zero). Its product of increasing bin weights is therefore at least the product of one copy of every bin weight. This last observation is why unequal shifted band masses cause only a controlled correction.

If $A/B' \le 2^{J'}$, applying (4.3) and the elementary bound $\log(n!) \le (n+1)\log n - n + 1$ gives the explicit prefix lower bound

$$\begin{aligned}\frac{\exp\{-nJ(\lambda) - 2\lambda C_0/(A+B') - J'\log(1+A/B') - 2 - 2n\xi(B')\}}{en^2} \\ - \frac{8n^2(A+B')}{B'\log 2}\, e^{-(A+B')}.\end{aligned} \tag{4.10}$$

For the product correction we used

$$\sum_{j<n} \log(1+(A/B')2^{-j}) \le J'\log(1+A/B') + 2.$$

The last term in (4.10) removes repeated prime marks. For the present parameters it is at most half the preceding main term, uniformly in all blocks. One can check this by comparing with the first block and the larger length k: the duplicate error is at most a polynomial in k times e^{-A}, whereas the main term is $e^{-O(k+\sqrt{k}\log k)}$.

For a suffix band with label ℓ, the exact probability of selecting one prime in its early subwindow is at least

$$\exp\left\{-\lambda\left(\log 2 + \frac{2C_0}{A+B2^{\ell}}\right)\right\}\frac{\lambda\eta/2}{1+(A/B)2^{-\ell}}. \tag{4.11}$$

Indeed its empty-selection probability is $e^{-\mu}$, its singleton odds sum is at least the intensity sum, and (4.5) bounds the latter below by $\lambda\eta B2^{\ell}/(2(A+B2^{\ell}))$ once $4C_0 \le \eta B$.

All blocks and their component bands are disjoint, so the probabilities multiply exactly. Use $J = 10h$ and $J_i = (J-ih)_+$ for block i. The shift corrections satisfy

$$\begin{aligned}\sum_{i<b} J_i \log(1+A/(B2^{ih})) &\le \min(b, \lfloor J/h\rfloor + 1)J(J+1)\log 2, \\ \sum_{\ell<k} \log(1+(A/B)2^{-\ell}) &\le (J+1)^2\log 2 + 2, \\ \sum_{i<b}\frac{1}{A+B2^{ih}} \le \frac{2}{B}, &\qquad \sum_{i<b}\xi(B2^{ih}) \le 2\xi(B).\end{aligned}$$

Thus the total negative logarithm is at most

$$bnJ(\lambda) + bK\big(\lambda \log 2 - \log(\lambda\eta/2)\big) + b(3 + \log 2 + 2\log n) \tag{4.12}$$
$$+ \min(b, \lfloor J/h \rfloor + 1)J(J+1)\log 2 + (J+1)^2 \log 2 + 2 + \frac{8\lambda C_0}{B} + 4n\xi(B).$$

Division by $k = bh$ gives $J(\lambda) + o(1)$, because $K/h \to 0$, $\log n/h \to 0$, and $h^2/k \to 0$. In particular the head-shift cost is paid only in a bounded number of early blocks.

For the deterministic estimates, the prefix of a block contributes at most $nA + nB'2^n$, and its suffix contributes at most $KA + B'e^\eta \sum_{j<K} 2^{n+j}$. Since $n \le (n2^{-K})2^K$, the block total is at most $hA + B'(e^\eta + n2^{-K})2^h$. Summing the geometric progression over completed blocks gives (4.8). The ballot condition and the singleton suffixes imply that the first j bands of the whole construction contain at most j primes. If p_{i+1} lies in band j, then $i \le j$, proving (4.9). □

For a finite set of primes $\mathcal{K}$, call its divisor cube H-proper if

$$(c_d)_{d|N(\mathcal{K})} \longmapsto \sum_{d|N(\mathcal{K})} c_d d \quad \text{is injective on } \{-H, \dots, H\}^{\{d : d|N(\mathcal{K})\}}. \tag{4.13}$$

Lemma 4.3 (Arithmetic removal of improper cores)**.** *The event in Lemma 4.2 may additionally be required to have an H-proper divisor cube, without changing its exponential rate. Every divisor subcube obtained from a subset of its primes is then H-proper.*

Proof. Suppose a divisor vector w is proper, but the doubled vector (w, pw) is not. Subtract two colliding coefficient vectors. The resulting equation is $U + pV = 0$, where U, V are old-cube differences with coefficient bound $2H$. Properness excludes $U = 0$ unless both old differences vanish. Consequently p divides a positive number in a set of size at most $(4H+1)^N$, where N is the old dimension. Each such number is at most $2H \sum w_i$.

For an old prime set S, $N = 2^{|S|}$ and $\sum w_i = \prod_{p\in S}(1+p)$. For any positive integer d, the number of its prime divisors exceeding e^v is at most $\log d/v$. Since $z_p \le 2/p$, the total selection weight of forbidden new primes with $\log p \ge v$ is at most

$$2e^{-v}(4H+1)^{2^{|S|}} \frac{\log(2H) + 2kX}{v}.$$

Take $v = A + B2^{|S|}$. For large k, $\log(4H+1) \le B$, so this is at most

$$\frac{2e^{-A}}{A}\big(\log(2H) + 2kX\big). \tag{4.14}$$

An improper selected core has a minimal improper subset. Removing its largest prime leaves a proper cube, and (4.9) still gives the required lower bound for that prime, since the smaller subset has no more preceding primes than the full core. This supplies a first-failure certificate.

For completeness, summing over all possible old subsets does not cost their unweighted number. Under independent selection, $\mathbb{P}(S \cup \{p\} \subseteq T) = z_p \prod_{q\in S} z_q$, and

$$\sum_{S\subseteq P} \prod_{q\in S} z_q = \prod_{q\in P}(1+z_q) \le \exp\left(\sum_{q\in P} z_q\right) \le e^{\lambda k \log 2 + O(1)}.$$

Multiplying this by (4.14) bounds the total failure probability by $\exp(-10k^3 + O(k))$. It is negligible compared with the lower bound in Lemma 4.2. Finally, extend coefficients on a subcube by zero on the other divisors to prove hereditary properness. □

For $t = ih$, $5 \le i \le b$, let

$$W_t = \prod_{p\in\mathcal{K}_t}(1+p), \qquad M_t = (2H+1)^{2^t}.$$

These are, respectively, the sum of the positive divisor weights and the cardinality of the proper signed cube at the anchor t. The endpoint estimate implies, uniformly over all successful cores,

$$YW_t \le M_t^{5/4}, \qquad 5h \le t \le k, \quad h \mid t. \tag{4.15}$$

Here is a useful explicit check on the uniformity. Since $\log(1+p) \le \log p + e^{-A}$, the additive remainder in $\log(YW_t)$ is at most $k^3 + 1 + t(A+1)$. For $t \ge 5h$ it is bounded by

$$\frac{100h^2}{2^h}\, k^2 2^t.$$

The coefficient of $k^2 2^t$ in the principal term tends to $(501/500)e^{\eta} < 5/4$. This proves (4.15). It also gives the strict inequality

$$\log N(\mathcal{K}) < \tfrac{5}{4} \log M_k, \tag{4.16}$$

because $N(\mathcal{K}) \le W_k$ and $Y > 1$. The first anchor is $5h$, and it satisfies the additional estimate

$$\frac{\log M_{5h}}{m \log Y} \le \frac{9}{2^{2h}} \longrightarrow 0. \tag{4.17}$$

The size condition and this last condition are both needed below.

4.3. **Positive logarithmic support under a supercritical tilt.** For a prime set T and $d > 0$, write

$$\mathcal{L}_d(T) = \bigcup_{S \subseteq T} \left[\sum_{p \in S} \log p, \sum_{p \in S} \log p + d \right].$$

We need the following finite support statement, uniformly in the number of dyadic bands.

Lemma 4.4 (Finite support estimate)**.** *Suppose $\lambda \log 2 > 1$ and $0 < b_0 \le B_0 < \infty$ are fixed. There are $c, C, a_0 > 0$ such that for $a \ge a_0$, $l \ge 1$, and $b_0 a \le d \le B_0 a$, a tilted selection of primes in $a < \log p \le a2^l$ satisfies*

$$\mathbb{P}_\lambda\big(|\mathcal{L}_d(T) \cap [0, Ca2^l]| \ge ca2^l\big) \ge \tfrac{1}{2}.$$

Proof. Let N_j be the number of primes in the band $a2^j < \log p \le a2^{j+1}$. The mean count tends uniformly to $\lambda \log 2 > 1$ as $a \to \infty$. Choose $1 < v < \lambda \log 2$. An exponential Markov bound for independent Bernoulli sums, followed by summation over suffix lengths, gives, with probability as close to one as desired,

$$\sum_{j=l-r}^{l-1} N_j \ge vr - D \qquad (0 \le r \le l) \tag{4.18}$$

for a fixed sufficiently large D. To justify the bound uniformly, choose $\theta > 0$ so small that $\theta v + (e^{-\theta} - 1)(\lambda \log 2 - o(1)) < 0$; the probability of failure at length r is at most $e^{-\theta D - cr}$. Likewise the expectation of

$$\sum_{j<l} N_j^2 e^{-a2^j}$$

tends to zero uniformly in l, using $\mathbb{E}N_j^2 \le (\mathbb{E}N_j)^2 + \mathbb{E}N_j$. Finally $\mathbb{E}\sum_{p \in T} \log p = O(a2^l)$; Markov's inequality provides a fixed upper bound on the support's location, with an arbitrarily small failure probability. This location event will be imposed only after the count conditioning has been removed.

Condition on a count vector satisfying the reverse barrier and the quadratic count regularity condition. Start with independent reciprocal-prime marks in each band: a mark has mass proportional to $1/p$. Equation (4.4) gives the small-ball bound

$$\mathbb{P}(|\log p - u| \le d) \le C_1 \frac{d+1}{a2^j}$$

in band j, uniformly in u. For two independent fair-bit subsets of the marked coordinates, partition their collision event according to the highest band in which their bits differ. Agreement above band j has

probability $2^{-N_{>j}}$, where $N_{>j} = \sum_{i>j} N_i$. Once the bits and all but one differing mark in band j are fixed, that last mark enters the difference with coefficient 1 or -1. The conditional collision probability is therefore bounded by the displayed small-ball estimate. Complete bit equality contributes 2^{-N} separately. Hence

$$\mathbb{P}\left(\left|\sum_i (\epsilon_i - \epsilon_i')\log p_i\right| \le d\right) \le 2^{-N} + C_1(d+1)\sum_{j<l}\frac{2^{-N_{>j}}}{a2^j}. \tag{4.19}$$

The reverse barrier makes the sum geometric. With $v = 1+\gamma$, it is at most $2^{D+1}/(a2^l(1-2^{-\gamma}))$, and $2^{-N} \le 2^D 2^{-l}$.

For fixed marks, convolve the fair-bit subset-sum measure with the uniform law on $[0,d]$, obtaining a density f of integral one. Write $\mathcal{S}$ for its support: this is the union of thickened subset sums of the indexed marks, including each coordinate separately. Then

$$\int f^2 \le \frac{1}{d}\mathbb{P}_{\epsilon,\epsilon'}(|S_\epsilon - S_{\epsilon'}| \le d).$$

Thus the mean energy is $O((a2^l)^{-1})$. Cauchy–Schwarz gives $1 \le |\mathcal{S}|\int f^2$, and Markov's inequality bounds the probability that $|\mathcal{S}| < ca2^l$ by an arbitrarily small fixed constant if $c > 0$ is chosen sufficiently small. We do not identify $\mathcal{S}$ with $\mathcal{L}_d(T)$ when the sampled tuple contains repeated primes.

It remains to pass from the auxiliary independent marks to the actual conditional subset law. On tuples distinct within each band, the latter is the reciprocal-prime product weight multiplied by $\prod p/(p-1)$, and then normalized. On these distinct tuples, $\mathcal{S} = \mathcal{L}_d(T)$. The quadratic count regularity above bounds both the repeated-mark loss and this product distortion. Choosing its fixed cutoff sufficiently small bounds the full conditional expectation of a nonnegative observable by twice its independent-mark expectation, simultaneously over all bands. Apply this to the small-support indicator, extended using the indexed support on tuples with repeats. There is one factor two for the entire tuple. Remove the conditioning on the counts and finally subtract the location failure probability. Small fixed error allowances for the two count conditions and for the location event yield probability at least $1/2$. □

Apply this lemma to a dyadic subwindow of $A < \log p \le X$, with $d_0 = A/1000 = k^3/100$. By monotonicity under enlarging the prime universe, there are constants $c > 0$ and $C \ge 100$, depending on λ, such that the support event has probability at least $1/2$ with support measure cX in $[0, CX]$. The event "contains a proper block core" is increasing, even though the event that the whole set is a core is not. Harris's inequality therefore combines the core and support events. The mean number of primes in the base window is at most $6k$. The exponential Markov estimate

$$\mathbb{P}(|T| > 18k) \le \exp\big((e-1)6k - 18k\big)$$

is negligible compared with the core probability. We have proved that, with probability at least $e^{-(J(\lambda)+o(1))k}$, the actual base set T_0 contains a proper core $\mathcal{K}$, has $|T_0| \le 18k$, and satisfies

$$|\mathcal{L}_{d_0}(T_0) \cap [0, CX]| \ge cX. \tag{4.20}$$

The core witness is retained as a subset of the original window $(A+B, X]$, so its final prefix is exactly the core used in (4.16).

4.4. **Two stages of prime-pair filling.** Put $L = CX$, choose $0 < \rho \le 1/2$ with $\rho C \le c$, and let r_0 be a fixed integer such that $(1-\rho/32)^{r_0} \le 2^{-10}$. Set $R = 8h$ and $e_0 = 32C_0$. We use $r_0 + R$ disjoint prime windows, with logarithmic starting points

$$u_i = 100(C+1)(i+1)X.$$

Their widths are $2L$ for $i < r_0$ and $w_j = L/2^{j+4}$ for $i = r_0 + j$, $0 \le j < R$. These windows lie above the entire base window.

We describe the geometric estimates and their transfer to actual prime pairs. The continuous comparison law for a logarithmic coordinate in $[u, u+w]$ has density

$$\frac{1}{x\log(1+w/u)}\,\mathbf{1}_{[u,u+w]}(x).$$

For two independent coordinates, their difference g has density $f_{u,w}$, supported on $[-w, w]$. When $w \le u$, $f_{u,w} \le 2/w$ everywhere. If $w = 2L$, it also satisfies $f_{u,2L}(g) \ge 1/(16L)$ for $|g| \le L$. The normalized intensity law a_p/μ on the prime window can be coupled to this continuous law with displacement at most $16C_0$. Indeed (4.5) bounds their distribution functions by $8C_0/w$, while the comparison quantile has Lipschitz constant at most $2w$. The pair differences can consequently be coupled with displacement at most e_0.

Lemma 4.5 (Initial and quadratic filling)**.** *Let T_0 be a base prime set with $|T_0| \le 18k$, and let $B_0 \subseteq [0, L] \cap \mathcal{L}_{d_0}(T_0)$ be compact with measure at least ρL. The actual tilted selections in these windows have, with probability $e^{-o(k)}$, exactly two primes in every window and choices of one representative q_i from every selected pair such that the selected prime product n_0 satisfies*

$$\left|[V + L/8, V + 7L/8] \setminus \bigcup_{d\mid n_0} [\log d, \log d + d_1]\right| \le \frac{L}{2^{3\cdot 2^R+R+7}}, \tag{4.21}$$

where

$$V = \sum_{i<r_0+R} \log q_i + (R+1)e_0, \qquad d_1 = d_0 + 2(R+1)e_0.$$

Here B_0 may depend on the observed base configuration, and n_0 is the product of the base and selected pair primes. The past prime set has at most $18k + 2(r_0 + R)$ elements.

Proof. Let $E \subseteq [0, L]$ be the currently uncovered set. Fubini's theorem and the density lower bound give

$$\mathbb{E}|E \setminus (B_0 + g)| \le (1 - \rho/16)|E|.$$

Consequently a fraction at least $\rho/32$ of shifts reduce the missing measure by the factor $1 - \rho/32$. After r_0 such successful steps, the uncovered fraction is at most 2^{-10}. At this first stage we retain the union of individual translates of B_0; the coupling error requires one common thickening, rather than an error accumulated over the number of initial translates.

For the second stage let A' be the current covered set in an interval $I = [a, b]$, with missing measure $E = |I \setminus A'|$. On the trimmed interval $I' = [a + w, b - w]$, the density upper bound and Fubini yield

$$\mathbb{E}|I' \setminus (A' \cup (A' + g))| \le 2E^2/w.$$

Thus at least half the shifts leave at most $4E^2/w$ uncovered. This also holds when $E = 0$, since a nonnegative function of integral zero vanishes almost everywhere. With $w = w_j = L/2^{j+4}$, the normalized missing measures satisfy

$$z_{j+1} \le 2^{j+6} z_j^2, \qquad z_0 \le 2^{-10}, \quad \Longrightarrow \quad z_j \le 2^{-(3\cdot 2^j+j+7)}.$$

The sum of the trimmings is less than $L/8$.

If two shifts differ by at most e_0, thickening the covered set by e_0 transfers its coverage to the displaced shift. Applying the quantile coupling therefore preserves each of these successful geometric inequalities for an actual prime pair, with the stated thickening. Repeated primes have probability at most $64e^{-u}/(w/u) \le 64/w$. All our widths, including the last one, tend to infinity, so the success probability after removing repeats is at least $\rho/64$ initially and at least $1/4$ subsequently. Equation (4.3), now with two marks, gives the actual per-window lower bound

$$\frac{\gamma}{32}\,\varepsilon^2 e^{-5\varepsilon}, \qquad \varepsilon = w/u, \qquad \gamma \in \{\rho/64, 1/4\}.$$

The negative logarithm of the product of these costs is $O((r_0+R+1)^2) = o(k)$, since the logarithm of the reciprocal width ratio in step j is $O(j+\log(r_0+R+1))$.

To recover actual divisors, write a pair as $\{p_i, q_i\}$ and use $g_i = \log p_i - \log q_i$. Adding the baseline $\sum_i \log q_i$ to a sum of selected differences replaces each q_i by either q_i or p_i. These are products of distinct primes in disjoint windows, multiplied by a divisor from the base set. Thus all translated atoms used above are logarithms of actual divisors. The symmetric thickening is converted to a one-sided interval by adding $(R+1)e_0$ to the origin. This gives (4.21). Conditional success at each step is uniform in the previous compact set; finite product conditioning multiplies the displayed costs. □

4.5. **Proper cubes, rational grids, and the high-prime window.** For positive integer weights $w = (w_1, \dots, w_N)$, set

$$F_w = \left\{ \alpha \in \mathbb{R}/\mathbb{Z} : \sum_i \|w_i\alpha\|^2 \le Y^{-2} \right\}, \qquad W = \sum_i w_i,$$

where $\|x\|$ is distance to the nearest integer.

Lemma 4.6 (Proper-cube grid bound)**.** *If the cube with coefficient bound H is proper and $Y \ge 32H$, put $M = (2H+1)^N$ and $R_w = YW$. If $R_w \ge 2$, then for every integer $q > 0$ the set*

$$S_q(w) = \{a \in \{0, \dots, q-1\} : \operatorname{dist}(a/q, F_w) \le 1/q\}$$

has cardinality at most $6(q+R_w)/M$.

Proof. Let C be the signed cube's set of integer values and let $\Phi(\alpha) = M^{-1}\sum_{x\in C} e^{2\pi i x\alpha}$. Properness gives $|C| = M$, hence exact character orthogonality gives $\int_{\mathbb{R}/\mathbb{Z}} |\Phi|^2 = 1/M$. For independent uniform coefficients $c_i \in \{-H, \dots, H\}$, the cross terms in their second moment vanish and

$$\mathbb{E}\left(\sum_i c_i t_i\right)^2 \le H^2 \sum_i t_i^2.$$

Using representatives t_i of $w_i\alpha$ of least absolute value, and $\cos x \ge 1 - x^2/2$, we obtain

$$\operatorname{Re}\Phi(\alpha) \ge 1 - 2\pi^2 H^2 \sum_i \|w_i\alpha\|^2.$$

If $\operatorname{dist}(\alpha,\beta) \le 1/(YW)$ and $\alpha \in F_w$, the triangle inequality gives $\sum_i \|w_i\beta\|^2 \le 4/Y^2$. Thus $|\Phi(\beta)| \ge 3/4$, and the $1/(YW)$-neighborhood of F_w has Haar measure at most $16/(9M)$. A finite grid packing converts this into the claimed count. Put $d = 1/(YW)$, and choose for each counted residue a witness in F_w within $1/q$. A maximal d-separated subfamily of these finitely many witnesses has disjoint open arcs of radius $d/2$, so its size times d is at most the measure of the d-neighborhood of F_w. Every other witness lies within d of a selected one. The corresponding grid points lie within $d + 1/q$ of that witness, and there are at most $2q(d+1/q)+1 = 2qd+3$ of them. Hence the total count is at most $(2q+3/d)$ times the neighborhood measure. This is at most $3(q+YW)16/(9M) < 6(q+YW)/M$. The argument includes all residues, not only residues coprime to q. □

Choose the final window after observing the pair representatives. Put

$$M = M_k, \quad \Delta = \frac{\log M}{8}, \quad u = V + L/4, \quad v = u - \Delta, \quad \varepsilon = \Delta/v, \quad U = \lceil e^u \rceil. \tag{4.22}$$

We select high primes from $Q = \{p : v < \log p \le u\}$. Uniformly in the successful past, the window is disjoint from the past prime universe and eventually satisfies

$$2\log M \le v \le \frac{m\log M}{16}, \qquad v \le e^k, \qquad k^{-1} \le \varepsilon \le 1. \tag{4.23}$$

These estimates follow directly by summing the bounds for $\log q_i \in (u_i, u_i + w_i]$: both u and v are of order $(r_0 + R + 1)^2 X$, while X is within fixed factors of $\log M$ and $(r_0 + R + 1)^2 = o(m)$. The same calculation, with $C \geq 100$, yields the margin

$$\log U + \tfrac{5}{4} \log M \leq V + L/2. \tag{4.24}$$

Lemma 4.7 (High-prime phase localization)**.** *Uniformly in every successful proper core and pair past, the actual tilted selection in Q has probability at least $e^{-o(k)}$ of consisting of exactly m primes, forming a set S such that*

$$\alpha \in F_{\mathcal{K}}, \quad p\alpha \in F_{\mathcal{K}} \ (p \in S) \quad \Longrightarrow \quad \|\alpha\| \leq U^{-1}. \tag{4.25}$$

Here $F_{\mathcal{K}}$ uses all the distinct divisors of $N(\mathcal{K})$ as its weights.

Proof. First use independent intensity marks in Q. The elementary progression bound obtained by majorizing prime reciprocals by reciprocals of all integers in the progression, together with (4.5), gives

$$\mathbb{P}(p \equiv a \pmod q) \leq \frac{G}{q}\left(1 + \frac{q}{P_{\min}}\right), \qquad G = 16(v + \varepsilon^{-1}), \quad P_{\min} = e^v. \tag{4.26}$$

It is enough that $8C_0 \leq \varepsilon v$, which holds here.

Suppose (4.25) fails. Reduced Dirichlet approximation gives $1 \leq q \leq U$, $(a, q) = 1$, and $\operatorname{dist}(\alpha, a/q) \leq 1/(qU)$. The case $q = 1$ implies $\|\alpha\| \leq U^{-1}$ and is excluded. Since the divisor weight 1 occurs, $\|\alpha\| \leq Y^{-1}$. For $q \geq 2$ the coprime grid point has norm at least $1/q$, so $\|\alpha\| \geq (1 - 1/U)/q$. Since $U > 2$, this excludes $q \leq Y/2$. For every mark $p \leq U$, multiplication on the circle shows that a/q and $(pa \bmod q)/q$ are each within $1/q$ of the relevant small-phase set. Reducedness permits multiplication by a to be cancelled in the residue count.

Write $E = M_{5h}$. For $Y/2 < q \leq E$, it suffices to use the weight 1. At most $2q/Y + 3 \leq 8q/Y$ residues have norm at most $Y^{-1} + q^{-1}$. Since $E \leq P_{\min}$, (4.26), independence of the m marks, and a union bound over the at most E^2 fractions give

$$E^2(16G/Y)^m. \tag{4.27}$$

For the remaining denominators use blocks $Q_0 \leq q \leq 2Q_0$ with $Q_0 = E2^j$. Select the largest anchor t for which $M_t \leq Q_0$, or the final anchor if $Q_0 \geq M$. Every full-core small phase is a small phase of the selected anchor. By Lemma 4.6 the old numerator and each marked numerator have at most $6(q+R_t)/M_t$ possibilities, where $R_t = YW_t \leq M_t^{5/4}$. For any one reduced old numerator, (4.26) bounds the probability that a mark passes by the residue count times its largest residue mass. Multiplying the m independent bounds and summing over old numerators and denominators gives

$$\frac{12(6G)^m(1 + 2Q_0/P_{\min})^m(2Q_0 + R_t)^{m+1}}{M_t^{m+1}Q_0^{m-1}}. \tag{4.28}$$

Here is the exponent calculation controlling all scales. If $Q_0 \leq P_{\min}$, write $Q_0 = M_t^a$. Consecutive anchors satisfy $M_{t+h} = M_t^{2^h}$, so $1 \leq a \leq 2^h$, except at the final anchor. Apart from an absolute constant to the power $m + 1$, the exponent of M_t in (4.28) is

$$(m + 1)\max(a, 5/4) - (m + 1) - (m - 1)a.$$

It is at most $-m/2$ when $m \geq 8\max(a, 1)$. At the final anchor let $a_U = \log U/\log M$. The window identities give

$$U \leq M^{a_U}, \quad P_{\min} \geq M^{5/4}, \quad P_{\min} \geq \frac{U}{2M^{1/8}}, \quad m \geq 8a_U.$$

For $Q_0 > P_{\min}$ the remaining factor is bounded by a constant to the power $m + 1$ times $M^{2a_U - 1 - 7m/8} \leq M^{-m/2}$. There are at most $1 + \log U/\log 2$ dyadic blocks. Thus the total independent-mark failure probability is at most

$$E^2(16G/Y)^m + \left(1 + \frac{\log U}{\log 2}\right) 1000^{m+1} G^m E^{-m/2}. \tag{4.29}$$

The constants deliberately leave room in the elementary inequalities. Uniformly under (4.23), $G \le e^{2k}$ and $1+\log U/\log 2 \le e^{2k}$ eventually. Equation (4.17) controls the first term, and $\log E \gg k$ controls the second. Each is at most e^{-k} for sufficiently large s.

Finally μ_Q lies between $\varepsilon/4$ and 5ε, and the repeated-mark probability is at most $m^2\, 16e^{-v}/\varepsilon \le e^{-k}$. Equation (4.3) therefore makes the actual good m-prime mass at least

$$\tfrac{1}{2}\exp\{-5\varepsilon - m\log(4m/\varepsilon)\} = e^{-o(k)},$$

because $m = O(k^{3/4})$ and $\varepsilon \ge k^{-1}$. All estimates were conditional on an arbitrary successful past. The exact identity (4.6) applies to the chosen fresh window, completing the proof without a union bound over past configurations. □

From now on n_0 denotes the product of the past primes and the selected high primes. Enlarging the prime set preserves the divisor-support estimate (4.21). The distinctness of the eventual coins is important. Set $b_{\mathcal{K}} = N(\mathcal{K})$. The high primes are fresh relative to $b_{\mathcal{K}}$, and the numbers

$$\{d : d \mid b_{\mathcal{K}}\} \cup \{pd : p \in S,\ d \mid b_{\mathcal{K}}\}$$

are all distinct: a collision between different branches would force a fresh prime to divide $b_{\mathcal{K}}$ or another high prime. They divide the selected rough product n_0. By (4.16) and (4.24), they are strictly smaller than every divisor $r \mid n_0$ with $\log r \ge V + L/2$. Restriction of a nonnegative phase sum to these coins now proves

$$\sum_{d\mid n_0,\ d<r} \|d\alpha\|^2 \le Y^{-2} \quad\Longrightarrow\quad \|\alpha\| e^{x_0} \le \tfrac{1}{2}, \qquad x_0 = \log(U/2). \tag{4.30}$$

4.6. **From logarithmic holes to sums of distinct divisors.** The final rough set has at most $20k$ primes eventually, since $2(r_0+R)+m = o(k)$. Hence n_0 is squarefree and has at most 2^{20k} divisors. Put

$$\omega = k^3/100, \quad g = d_1 + \omega, \quad a = V + L/2, \quad b = V + 3L/4.$$

For large s, the following explicit inequalities hold:

$$\begin{gathered} g \le k^3/20, \qquad e^{2g} \le Y/2, \qquad 2(g+\log 4) \le L/4, \\ V + L/8 + \omega + g \le x_0, \qquad L\,2^{-(3\cdot 2^R+R+7)} < \omega. \end{gathered} \tag{4.31}$$

The last inequality follows from $k/2^R \to 0$, since $R = 8h$ and $k = h2^{4h}$. Thus (4.21) has missing measure smaller than ω.

We include the deterministic conversion underlying the construction. If the complement of $\bigcup_{d\mid n_0}[\log d, \log d + d_1]$ in an interval has measure less than ω, then every admissible interval $[x-\omega, x]$ meets this union. Consequently an actual divisor satisfies $x - g \le \log d \le x$. This is a logarithmic mesh, not an assertion that the support itself is a full interval.

For $r \mid n_0$ with $a \le \log r \le b$, let $\mathcal{D}_r = \{d : d \mid n_0, d < r\}$. We claim that every nonzero frequency modulo r satisfies

$$\sum_{d\in\mathcal{D}_r} \|ad/r\|^2 > Y^{-2} \qquad (1 \le a < r). \tag{4.32}$$

If not, replace a/r by its signed representative x of least absolute value. Equation (4.30) gives $|x|e^{x_0} \le 1/2$, and each $|x|d$ is within Y^{-1} of an integer. For $d \le e^{x_0}$ this forces $|x|d \le Y^{-1}$. For a larger divisor $d \le r$, apply the mesh at $x = \log d - g$. The inequalities in (4.31) keep this query inside the interval of coverage. It supplies d' with $\log d - 2g \le \log d' \le \log d - g$, hence $d' < d$ and $d/d' \le e^{2g} \le Y/2$. Induction on the integer d shows $|x|d' \le Y^{-1}$, so $|x|d \le 1/2$; proximity to an integer again forces $|x|d \le Y^{-1}$. Include the endpoint coin r in this induction; its phase is exactly an integer, so its distance to the nearest integer is zero without needing a term for r in the assumed phase sum. But $|x|r$ is a nonzero integer, contradicting $Y > 2$.

Take a fair Boolean sum of the distinct coins $d \in \mathcal{D}_r$ modulo r. Its normalized Fourier transform at a has absolute value

$$\prod_{d\in\mathcal{D}_r} |\cos(\pi ad/r)| \le \exp\left(-2\sum_{d\in\mathcal{D}_r} \|ad/r\|^2\right) \le e^{-2/Y^2}.$$

With $L_r = \lceil Y^2 \log(2r)\rceil$ independent copies, the sum of the absolute values of the nontrivial Fourier coefficients is less than one. Finite Fourier inversion therefore gives positive mass to every residue modulo r. Each residue has a representative $\sum_{d\in\mathcal{D}_r} c_d d$ with $0 \le c_d \le L_r$.

Uniformly over the divisors in $[a, b]$, $\log(2r) \le e^{2k}$ eventually. Since $|\mathcal{D}_r| \le 2^{20k}$, our choice $Z = \lceil e^{3k^3}\rceil$ ensures

$$L_r \le Z, \qquad 4L_r \sum_{d\in\mathcal{D}_r} d \le Zr. \tag{4.33}$$

For any integer $t \in [Zr/4, 3Zr/4]$, choose the residue representative of t. It is at most $L_r \sum d \le t$, so adding a nonnegative multiple of the reserve coin r yields

$$t = br + \sum_{d\in\mathcal{D}_r} c_d d, \qquad 0 \le b \le Z, \quad 0 \le c_d \le Z. \tag{4.34}$$

Lemma 4.8 (A practical factor at negligible cost)**.** *In the prime coordinates below e^A, independently of the rough construction, an event of tilted probability $e^{-o(k)}$ supplies a practical divisor b_0 with $Z \le b_0 < Z^2$. It is coprime to the rough product. On this event every integer in $[Zr/4, 3Zr/4]$ is a sum of distinct divisors of $b_0 n_0$.*

Proof. Here is an elementary finite seed construction. Require $2, 3, 5, 7$, and one prime from each of $O(\log\log Z)$ consecutive windows

$$3(3/2)^j < \log p \le 4(3/2)^j.$$

Each window has a positive hit probability bounded below uniformly in j, by (4.4); the finitely many early windows are nonempty by elementary prime interval bounds. The joint event has probability at least $c/(\log Z)^{C_2}$ in the original model and at least this probability under $\lambda \ge 1$ by monotonicity. Starting at 210, multiply by one selected prime from each window. The preceding product is larger than the next prime: its logarithm is at least $\log 210 + 6((3/2)^j - 1)$, whereas the next logarithm is at most $4(3/2)^j$. Such extensions preserve practicality, as can be seen by overlapping the intervals of divisor sums contributed by the old divisors and their multiples by the new prime. Stop when the product first crosses Z; it is then smaller than Z^2. A deterministic number $O(\log\log Z)$ of windows suffices, all lying below Z. Since $\log\log Z = O(\log k)$, this costs $e^{-o(k)}$. All rough primes exceed $e^A > Z^2$, so coprimality and independence hold.

For each coefficient in (4.34), use practicality to express it as a sum of distinct divisors of b_0. Multiplying these representations by their respective distinct divisors of n_0 produces distinct divisors of $b_0 n_0$. Indeed coprimality makes the decomposition into a b_0-divisor and an n_0-divisor unique. This proves the last assertion. □

4.7. **Harmonic averaging and the final change of measure.** All selected primes may be viewed in the fixed universe

$$P_s = \{p : p \text{ prime},\ p \le \exp(k^3 X)\}.$$

Use the fixed target horizon

$$\mathcal{H}_s = \{t \in \mathbb{N} : e^X \le t \le e^{k^3 X}\},$$

with integer endpoints understood. Every interval $[Zr/4, 3Zr/4]$ used above lies in $\mathcal{H}_s$ for all sufficiently large s. The selected arithmetic product divides $N(T)$ for the ambient prime set T; its representations are consequently also representations by distinct divisors of $N(T)$.

Define a single bounded observable

$$F_s(T) = \frac{\displaystyle\sum_{t\in\mathcal{H}_s} \frac{1}{t}\mathbf{1}_{\{N(T) \text{ represents } t\}}}{\displaystyle\sum_{t\in\mathcal{H}_s} \frac{1}{t}}.$$

The mesh gives a quantitative lower bound on this observable. Choose mesh divisors at spacing $g+\log 4$ in logarithmic coordinates. After discarding endpoint losses there are at least $(b-a)/(2(g+\log 4))$ such divisors, separated by multiplicative ratio at least 4. Their target intervals are disjoint. For every integer $N \geq 4$, the integers in $[N/4, 3N/4]$ have total reciprocal weight at least $1/4$. Also $\sum_{t\leq e^{k^3X}} 1/t \leq 1+k^3X$. It follows that on every successful construction,

$$F_s(T) \geq \frac{L/4}{8(g+\log 4)(1+k^3X)} \geq \frac{1}{256k^6} = e^{-o(k)}. \tag{4.35}$$

The harmless final constant follows from $C \geq 100$ and $g \leq k^3/20$. This averaging step requires only a logarithmic mesh and does not claim positive coverage at every individual target.

Combining Lemmas 4.2–4.8, Harris's inequality for the base, exact independence of the seed, and (4.6) for the last window, we have established the quantitative statement

$$\mathbb{E}_{P_s,\lambda} F_s \geq \exp\bigl(-(J(\lambda)+o(1))k\bigr) \qquad (\lambda_* < \lambda < 2). \tag{4.36}$$

For clarity, we finish the change of measure explicitly. Under the original pattern law the likelihood is $L_\lambda(T) = \lambda^{|T|}/Z_{P_s}(\lambda)$, and its exact moment is

$$\mathbb{E}_{P_s,1} L_\lambda^q = \frac{Z_{P_s}(\lambda^q)}{Z_{P_s}(\lambda)^q}.$$

For any fixed $q > 1$, Mertens's estimate and the summable quadratic error in $\log(1+x)$ give

$$\log \mathbb{E}_{P_s,1} L_\lambda^q = \bigl(\lambda^q - 1 - q(\lambda-1) + o(1)\bigr)k\log 2. \tag{4.37}$$

Here $\log\log\exp(k^3X) = \log(k^3X) = k\log 2 + O(\log k)$. For each fixed t, the original-pattern probability that $N(T)$ represents t is at most $d(t)$, since representation persists in every multiple of $N(T)$. Thus $\mathbb{E}_{P_s,1} F_s$ is at most the harmonic average $\overline{d}_s$ of the actual densities over $\mathcal{H}_s$. Since $0 \leq F_s \leq 1$, one application of Hölder gives

$$\mathbb{E}_{P_s,\lambda} F_s \leq \overline{d}_s^{\,1-1/q} \left(\mathbb{E}_{P_s,1} L_\lambda^q\right)^{1/q}.$$

There is no likelihood penalty separately for the seed, the core, the pair windows, or the high primes. They belong to one finite observable on one prime universe.

Equations (4.36) and (4.37) show that some $t_s \in \mathcal{H}_s$ satisfies

$$\frac{-\log d(t_s)}{\log\log t_s} \leq \frac{qJ(\lambda)/\log 2 + \lambda^q - 1 - q(\lambda-1)}{q-1} + o(1). \tag{4.38}$$

Indeed a positive weighted average has an entry at least as large as the average, and $\log\log t \geq \log X \geq k\log 2$ throughout the horizon. Its lower endpoint tends to infinity.

Finally, for fixed $q > 1$ let $\lambda \downarrow \lambda_*$, so $J(\lambda) \to 0$, and then let $q \downarrow 1$. The right side tends to

$$\lambda_* \log \lambda_* - \lambda_* + 1 = 1 - \frac{1+\log\log 2}{\log 2} = \delta.$$

For each prescribed error, choose these two parameters first and then take s sufficiently large. This proves Theorem 4.1.

5. The critical exponent and the analytic contradiction

Set

$$\delta = 1 - \frac{1 + \log\log 2}{\log 2}, \qquad \lambda_* = \frac{1}{\log 2}. \tag{5.1}$$

One has $0 < \delta < 1$ and $1 < \lambda_* < 2$. For completeness, $1/2 < \log 2 < 1$, and strict concavity of the logarithm gives $\log\log 2 < \log 2 - 1$, proving $\delta > 0$. Also $\log\log 2 > -\log 2 > -1$, proving $\delta < 1$.

The two quantitative conclusions used here are

$$(\log t)^{\delta} d(t) \longrightarrow 0 \tag{5.2}$$

and

$$\liminf_{t\to\infty} \frac{-\log d(t)}{\log\log t} = \delta. \tag{5.3}$$

The first is an upper estimate for every sufficiently large target. The second requires lower bounds only along arbitrarily large targets. This distinction is useful because the lower construction proceeds through a sequence of finite target horizons.

Lemma 5.1 (A sufficient lower input)**.** *Suppose* (5.2) *holds and, for every $\varepsilon > 0$, arbitrarily large integers t satisfy*

$$d(t) \geq \exp\bigl(-(\delta + \varepsilon)\log\log t\bigr). \tag{5.4}$$

Then (5.3) *holds.*

Proof. By Proposition 2.1, $d(t) > 0$. For all sufficiently large t, (5.2) gives $(\log t)^{\delta} d(t) \leq 1$. Taking logarithms and dividing by the positive quantity $\log\log t$ yields

$$\frac{-\log d(t)}{\log\log t} \geq \delta.$$

On each of the arbitrarily large targets in (5.4), the reverse estimate with $\delta + \varepsilon$ holds. Consequently the liminf is at least δ and at most $\delta + \varepsilon$ for every $\varepsilon > 0$. It is therefore δ.

In particular this is a finite real liminf. The eventual lower bound supplies boundedness from below, while, for example, the infinitely often bound by $\delta + 1$ prevents the lower tail from escaping to $+\infty$. These are the boundedness conditions used in the formal real-valued filter liminf. □

Proposition 5.2 (Analytic obstruction to a power law)**.** *If* (5.2) *and* (5.3) *hold, then there are no constants $c > 0$ and $p \in \mathbb{R}$ such that*

$$d(t) \sim \frac{c}{(\log t)^p}.$$

Proof. Suppose otherwise and write

$$d(t) = \frac{c}{(\log t)^p}\,\phi(t), \qquad \phi(t) \longrightarrow 1.$$

For all sufficiently large t, the factors are positive. Taking logarithms gives

$$\frac{-\log d(t)}{\log\log t} = p - \frac{\log c + \log\phi(t)}{\log\log t} \longrightarrow p.$$

Its liminf is δ by (5.3), so $p = \delta$. The assumed equivalence would then give $(\log t)^{\delta} d(t) \longrightarrow c$, contradicting (5.2) and $c > 0$. □

Corollary 5.3 (The negative answer)**.** *There do not exist positive real constants c_1, c_2 and a function e assigning the natural density of $\mathcal{D}_t$ for each $t > 0$ such that*

$$e(t) \sim \frac{c_1}{(\log t)^{c_2}}.$$

Proof. Theorem 3.1 proves (5.2), and Theorem 4.1 together with Lemma 5.1 proves (5.3). Uniqueness of natural density and Proposition 2.1 give $e(t) = d(t)$ for all $t > 0$. The asserted equivalence is therefore excluded by Proposition 5.2. □

6. Formal verification and proof artifact

The companion file `erdos_859_counterexample.lean` contains the complete formal proof. It was checked with Lean 4.33.1 against Formal Conjectures commit

`8432eac998110a563e03df65a28c117e97c8c142.`

Its two imports are the pinned Erdős 859 problem file and the challenge support module `TaskSupport`. It contains 37,218 lines, including the supporting estimates, audit commands, and the applicable third-party license. The mathematical development was assembled from 253 development modules into a single source file.

The formal target is

```
theorem Bounty.target :
    ¬ (fcTypeOfName% "Erdos859.erdos_859")
```

The macro `fcTypeOfName%` retrieves the type of the named conjecture. The proof does not invoke the imported conjecture's placeholder proof. It also proves existence and positivity of the canonical density directly, without invoking placeholder variants in the problem file.

The final declaration applies the analytic contradiction to two proved theorems:

(1) `divisorDensity_log_rate_liminf`, which proves the lower-limit identity in Theorem 1.1;
(2) `divisorDensity_critical_scaled_tendsto_zero`, which proves the vanishing scaled-density limit.

Both declarations lie in the namespace `Erdos859Verification`. The final target has no further mathematical hypotheses.

6.1. Correspondence with the paper.

The following list identifies the principal formal endpoints. Intermediate finite identities, error bounds, and eventual parameter inequalities are proved in the same file.

Density and periodicity.:
`divisorSumSet_hasDensity` and the positivity results construct the canonical d_t for the exact divisor-sum set.

Divisor-interval upper bound.:
The `FordSupport`, `FordRadical`, and `FordPowerful` declarations prove the support estimate, stopping-prime reduction, and restoration of prime powers. The `DivisorDyadic` declarations transfer them to the density upper limit.

The tilted arithmetic construction.:
`corePastHigh_event_probability_lower` provides the event containing the proper core, positive base support, successful pair stages, and the fixed-count high-prime phase condition. Its proof conditions on the past before selecting the adaptive high window.

Actual integer representations.:
`coreArithmeticCoverage_exponential_eventually` supplies the harmonic coverage lower bound for the selected prime product and a practical seed. `coreSeedPastHigh_coverage_eventually` provides that seed inside the same global prime selection.

The change of measure.:
`coreGlobalHarmonicExpectation_lower` proves the tilted expectation bound. `coreGlobalHarmonicExpectation_frequent_rate` applies the single Hölder transfer and extracts arbitrarily large targets.

6.2. Audit and reproducibility.

The compiler completed with exit code zero with automatic implicit parameters disabled and warnings treated as errors. All 975 printed axiom audits use only the standard Lean axioms. In particular, the final audit is

```
'Bounty.target' depends on axioms:
    [propext, Classical.choice, Quot.sound].
```

The delivered source contains no unfinished proof, added axiom, or custom unsafe declaration. The supporting estimates are proved in the artifact; they are not parameters of the final theorem.

The SHA-256 digest of the checked file is

`bebc763e69acfecdbac6744e883eb44f26b5`
`60e11056c8d630201f01c4a112ee.`

The source archive includes that exact file, its verification log and manifest, and the challenge support file used for the check. The manifest records source hashes and the dependency revision. With Python 3.9 or later and an already compiled checkout of the pinned Formal Conjectures environment, run the supplied verifier from the companion directory:

```
python verify_lean.py --fc-root /path/to/formal-conjectures
  --lean /path/to/lean --log verification.log
  --manifest verification.json
```

Enter the command on one line; it is wrapped here for readability. The verifier checks the compiler version, source hashes, and repository revision, then compiles the original problem, challenge support, and complete proof in a fresh temporary overlay. It requires the final axiom audit above and does not change the dependency checkout. The accompanying instructions explain the dependency prerequisites and the optional report paths. The mathematical proof is contained in the single companion file; the imported Lean libraries and the pinned problem statement are its environment. As in an ordinary Lean build, the precompiled dependencies are part of the trusted environment.

The adapted prime-reciprocal development derives from PNT+ commit

`a5154676af9aa3095150ee410cdda80555aa0642.`

Its Apache 2.0 license and provenance are retained. We also acknowledge the Lean, Mathlib, and Formal Conjectures contributors whose libraries support the formal development.

Acknowledgments and AI assistance

We acknowledge extensive assistance from OpenAI's Codex in exploring the mathematical argument, developing and debugging the Lean formalization, and drafting and checking the exposition. The accompanying formal proof was checked by Lean's kernel as described in Section 6. We thank the contributors to the PNT+ project for the prime-reciprocal formalization used in the proof and the maintainers of the Formal Conjectures library for the precise problem statement.

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